U-Net: Convolutional Networks for Biomedical Image Segmentation

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Abstract. There is large consent that successful training of deep networks requires many thousand annotated training samples. In this paper, we present a network and training strategy that relies on the strong use of data augmentation to use the available annotated samples more efficiently. The architecture consists of a contracting path to capture context and a symmetric expanding path that enables precise localization. We show that such a network can be trained end-to-end from very few images and outperforms the prior best method (a sliding-window convolutional network) on the ISBI challenge for segmentation of neuronal structures in electron microscopic stacks. Using the same network trained on transmitted light microscopy images (phase contrast and DIC) we won the ISBI cell tracking challenge 2015 in these categories by a large margin. Moreover, the network is fast. Segmentation of a 512x512 image takes less than a second on a recent GPU. The full implementation (based on Caffe) and the trained networks are available at http://lmb.informatik.uni-freiburg.de/people/ronneber/u-net.

1 Introduction

In the last two years, deep convolutional networks have outperformed the state of the art in many visual recognition tasks, e.g. [7,3]. While convolutional networks have already existed for a long time [8], their success was limited due to the size of the available training sets and the size of the considered networks. The breakthrough by Krizhevsky et al. [7] was due to supervised training of a large network with 8 layers and millions of parameters on the ImageNet dataset with 1 million training images. Since then, even larger and deeper networks have been trained [12].

The typical use of convolutional networks is on classification tasks, where the output to an image is a single class label. However, in many visual tasks, especially in biomedical image processing, the desired output should include localization, i.e., a class label is supposed to be assigned to each pixel. Moreover, thousands of training images are usually beyond reach in biomedical tasks. Hence, Ciresan et al. [1] trained a network in a sliding-window setup to predict the class label of each pixel by providing a local region (patch) around that pixel

U-Net：用于生物医学图像分割的卷积网络

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摘要。广泛认为，成功训练深度网络需要数千个带注释的训练样本。本文提出了一种网络和训练策略，依靠大量数据增强，更高效地利用可用的注释样本。该架构由一条收缩路径用于捕捉上下文和一条对称扩展路径组成，实现精确的定位。我们证明，这种网络可以从极少的图像进行端到端训练，并且在电子显微镜堆栈中神经元结构分割的 ISBI 挑战中，表现优于之前的最佳方法（滑动窗口卷积网络）。使用同一网络，训练于透射光显微镜图像（相位差和 DIC），我们在 2015 年 ISBI 细胞追踪挑战赛中以大幅优势获胜。此外，网络速度很快。在最近的 GPU 上，512x512 图像的分割不到一秒。完整的实现（基于 Caffe）和训练后的网络可在 http://lmb.informatik.uni-freiburg.de/people/ronneber/u-net 获取。

1 简介

在过去两年里，深度卷积网络在许多视觉识别任务中表现优于最先进技术，例如[7,3]。虽然卷积网络早已存在[8]，但由于可用训练集的规模和所考虑的网络规模，其成功度有限。Krizhevsky 等人[7]的突破，得益于 ImageNet 数据集上对一个拥有 8 层和数百万参数的大型网络进行监督训练，该数据集包含 100 万张训练图像。此后，训练了更大更深的网络[12]。

卷积网络的典型用途是在分类任务中，其中图像输出为单一类别标签。然而，在许多视觉任务中，尤其是生物医学图像处理中，期望输出应包括定位，即每个像素应分配类别标签。此外，成千上万的训练图像通常无法在生物医学任务中获得。因此，Ciresan 等人[1]在滑动窗口设置中训练了一个网络，通过在像素周围提供局部区域（patch）来预测该像素的类标签

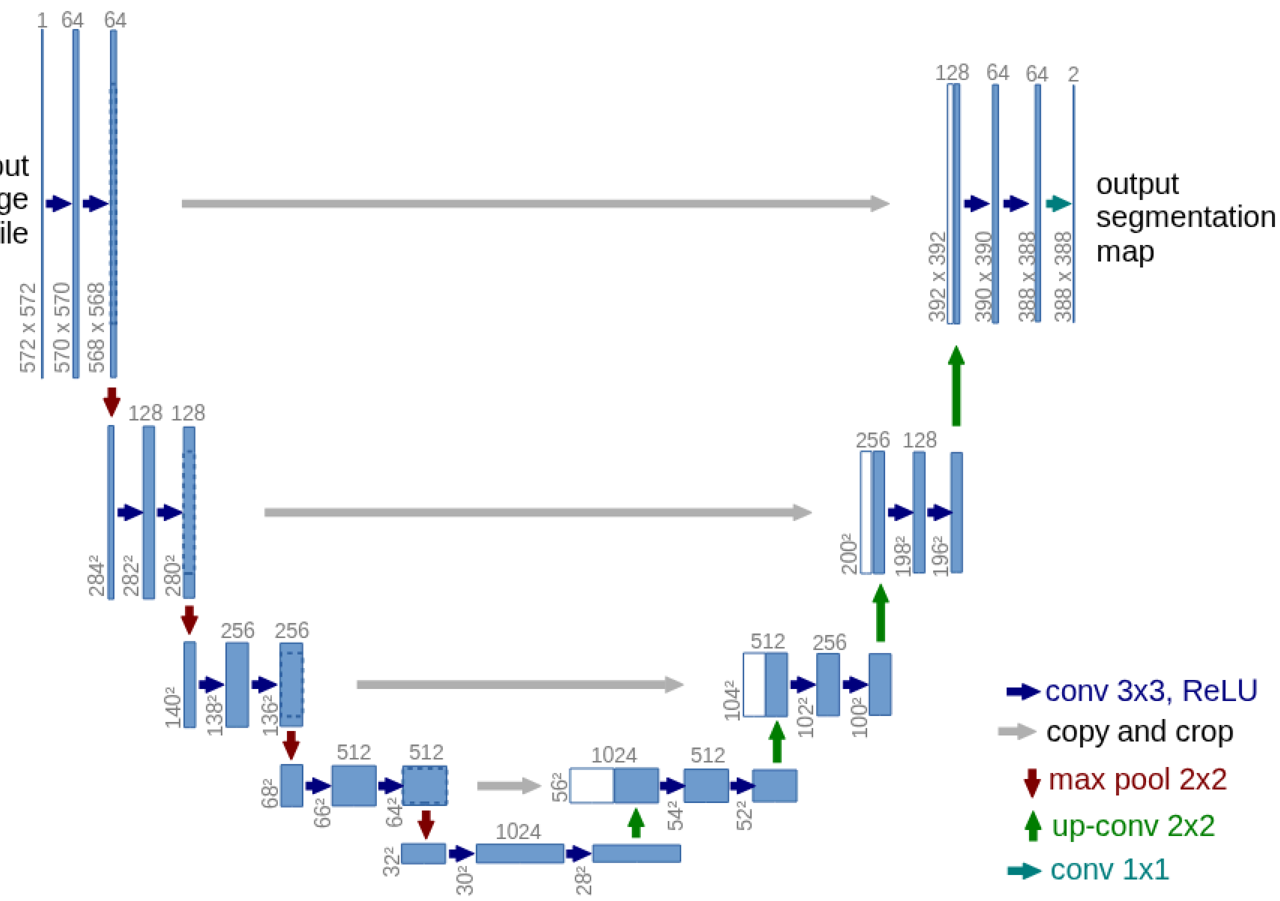


Fig. 1. U-net architecture (example for 32x32 pixels in the lowest resolution). Each blue box corresponds to a multi-channel feature map. The number of channels is denoted on top of the box. The x-y-size is provided at the lower left edge of the box. White boxes represent copied feature maps. The arrows denote the different operations.

as input. First, this network can localize. Secondly, the training data in terms of patches is much larger than the number of training images. The resulting network won the EM segmentation challenge at ISBI 2012 by a large margin.

Obviously, the strategy in Ciresan et al. [1] has two drawbacks. First, it is quite slow because the network must be run separately for each patch, and there is a lot of redundancy due to overlapping patches. Secondly, there is a trade-off between localization accuracy and the use of context. Larger patches require more max-pooling layers that reduce the localization accuracy, while small patches allow the network to see only little context. More recent approaches [11,4] proposed a classifier output that takes into account the features from multiple layers. Good localization and the use of context are possible at the same time.

In this paper, we build upon a more elegant architecture, the so-called "fully convolutional network" [9]. We modify and extend this architecture such that it works with very few training images and yields more precise segmentations; see Figure 1. The main idea in [9] is to supplement a usual contracting network by successive layers, where pooling operators are replaced by upsampling operators. Hence, these layers increase the resolution of the output. In order to localize, high resolution features from the contracting path are combined with the upsampled

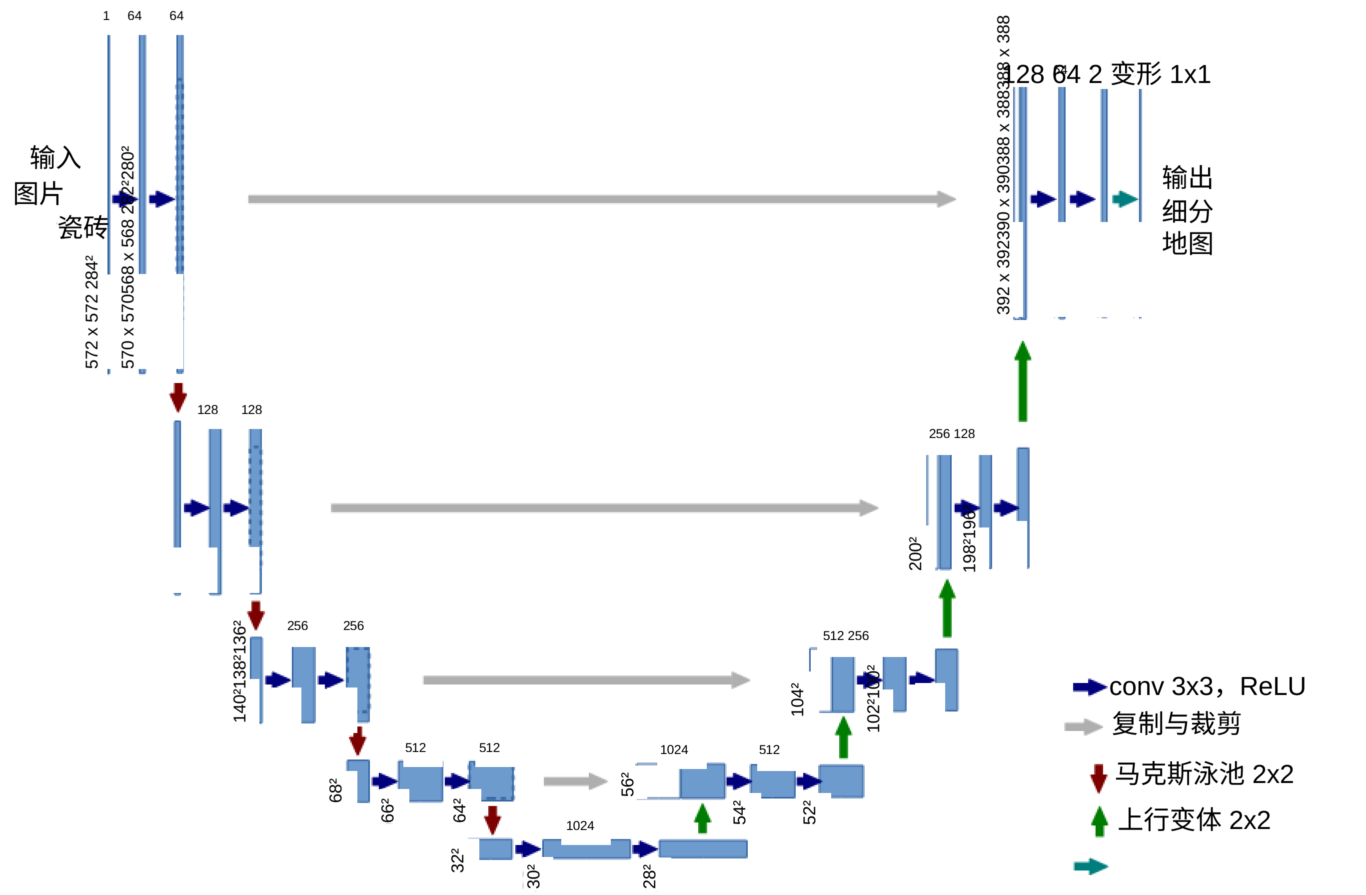


图 1。U-net 架构（最低分辨率下 32x32 像素的例子）。每个蓝色方框对应多通道特征图。信道数量标注在盒子顶部。x-y 尺寸显示在盒子的左下角。白色方框代表复制的特征图。箭头表示不同的操作。

作为输入。首先，这个网络可以本地化。其次，训练数据以补丁数计算，远大于训练图像的数量。最终的网络在 2012 年 ISBI 的电磁分段挑战中以大幅优势获胜。

显然，Ciresan 等人[1]的策略有两个缺点。首先，它相当慢，因为每个补丁都必须单独运行网络，而且由于补丁重叠，存在大量冗余。其次，定位精度与上下文使用之间存在权衡。较大的补丁需要更多的最大池层，降低定位准确性，而小补丁则使网络只能看到较少的上下文。较新的方法[11,4]提出了一种考虑多层特征的分类器输出。良好的定位和上下文的运用是可以同时实现的。

本文基于更优雅的架构——所谓的“全卷积网络”[9]。我们修改并扩展了该架构，使其只需使用极少的训练图像，并实现更精确的分割;见图 1。[9]中的主要思想是通过层叠加通常的合同网络来补充，其中池化算子被上采样算子取代。因此，这些层增加了输出的分辨率。为了定位，收缩路径中的高分辨率特征与上采样结合

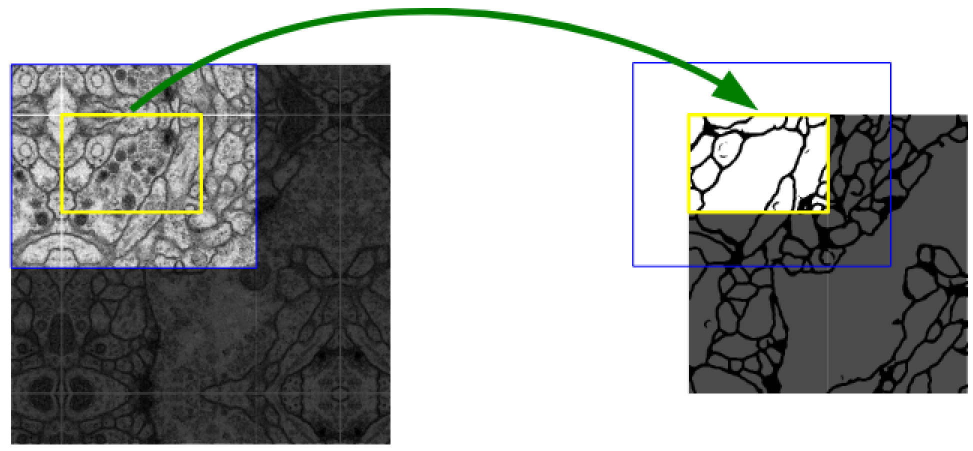


Fig. 2. Overlap-tile strategy for seamless segmentation of arbitrary large images (here segmentation of neuronal structures in EM stacks). Prediction of the segmentation in the yellow area, requires image data within the blue area as input. Missing input data is extrapolated by mirroring

output. A successive convolution layer can then learn to assemble a more precise output based on this information.

One important modification in our architecture is that in the upsampling part we have also a large number of feature channels, which allow the network to propagate context information to higher resolution layers. As a consequence, the expansive path is more or less symmetric to the contracting path, and yields a u-shaped architecture. The network does not have any fully connected layers and only uses the valid part of each convolution, i.e., the segmentation map only contains the pixels, for which the full context is available in the input image. This strategy allows the seamless segmentation of arbitrarily large images by an overlap-tile strategy (see Figure 2). To predict the pixels in the border region of the image, the missing context is extrapolated by mirroring the input image. This tiling strategy is important to apply the network to large images, since otherwise the resolution would be limited by the GPU memory.

As for our tasks there is very little training data available, we use excessive data augmentation by applying elastic deformations to the available training images. This allows the network to learn invariance to such deformations, without the need to see these transformations in the annotated image corpus. This is particularly important in biomedical segmentation, since deformation used to be the most common variation in tissue and realistic deformations can be simulated efficiently. The value of data augmentation for learning invariance has been shown in Dosovitskiy et al. [2] in the scope of unsupervised feature learning.

Another challenge in many cell segmentation tasks is the separation of touching objects of the same class; see Figure 3. To this end, we propose the use of a weighted loss, where the separating background labels between touching cells obtain a large weight in the loss function.

The resulting network is applicable to various biomedical segmentation problems. In this paper, we show results on the segmentation of neuronal structures in EM stacks (an ongoing competition started at ISBI 2012), where we out-

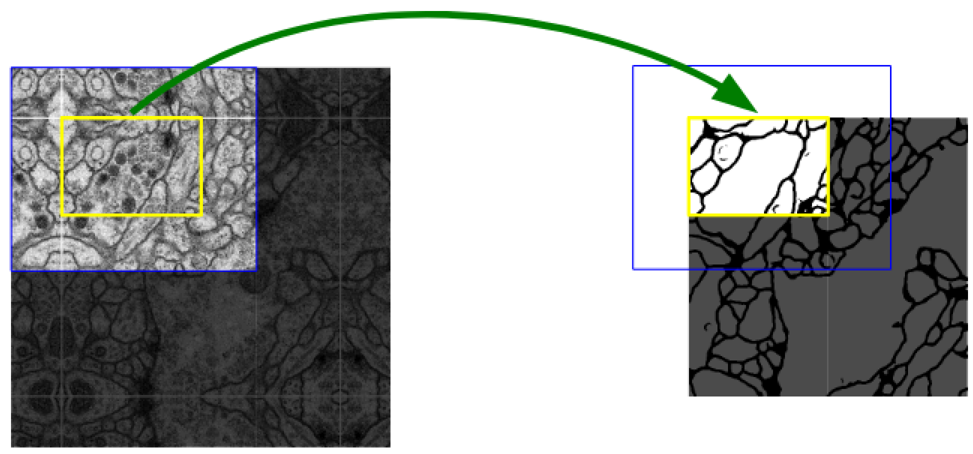


图 2。重叠瓦片策略用于无缝分割任意大型图像（此处为 EM 堆栈中神经结构的分割）。预测黄色区域的分割需要输入蓝色区域内的图像数据。缺失的输入数据通过镜像外推

输出。随后，连续的卷积层可以学习基于这些信息组装出更精确的输出。

我们架构中的一个重要修改是，在上采样部分我们还包含了大量特征通道，使网络能够将上下文信息传播到更高分辨率的层。因此，膨胀路径与收缩路径大致对称，形成 U 形架构。该网络没有完全连通的层，只使用每个卷积的有效部分，即分段映射只包含像素，输入图像中包含完整上下文。该策略允许通过重叠瓦片策略无缝分割任意大的图像（见图 2）。为了预测图像边界区域的像素，通过镜像输入图像外推缺失的上下文。这种镶嵌策略对于将网络应用于大型图像非常重要，否则分辨率会受限于 GPU 内存。

至于我们的任务，可用的训练数据非常有限，我们通过对现有训练图像施加弹性变形来进行大量数据增强。这使得网络能够学习对这些变形的不变性，而无需在注释图像语料库中看到这些变换。这在生物医学分割中尤为重要，因为变形曾是组织中最常见的变异，真实变形可以高效地模拟。数据增强对学习不变性的重要性已在 Dosovitskiy 等人[2]中被证明，适用于无监督特征学习。

许多单元格分割任务中的另一个挑战是分离同类接触对象;见图 3。为此，我们提出加权损失，即接触单元之间的背景标记在损失函数中获得较大权重。

由此产生的网络适用于各种生物医学分段问题。本文展示了 EM 堆栈中神经结构分割的研究成果（该竞赛始于 2012 年 ISBI 展会），我们超越了——

performed the network of Ciresan et al. [1]. Furthermore, we show results for cell segmentation in light microscopy images from the ISBI cell tracking challenge 2015. Here we won with a large margin on the two most challenging 2D transmitted light datasets.

2 Network Architecture

The network architecture is illustrated in Figure 1. It consists of a contracting path (left side) and an expansive path (right side). The contracting path follows the typical architecture of a convolutional network. It consists of the repeated application of two 3x3 convolutions (unpadded convolutions), each followed by a rectified linear unit (ReLU) and a 2x2 max pooling operation with stride 2 for downsampling. At each downsampling step we double the number of feature channels. Every step in the expansive path consists of an upsampling of the feature map followed by a 2x2 convolution ("up-convolution") that halves the number of feature channels, a concatenation with the correspondingly cropped feature map from the contracting path, and two 3x3 convolutions, each followed by a ReLU. The cropping is necessary due to the loss of border pixels in every convolution. At the final layer a 1x1 convolution is used to map each 64-component feature vector to the desired number of classes. In total the network has 23 convolutional layers.

To allow a seamless tiling of the output segmentation map (see Figure 2), it is important to select the input tile size such that all 2x2 max-pooling operations are applied to a layer with an even x- and y-size.

3 Training

The input images and their corresponding segmentation maps are used to train the network with the stochastic gradient descent implementation of Caffe [6]. Due to the unpadded convolutions, the output image is smaller than the input by a constant border width. To minimize the overhead and make maximum use of the GPU memory, we favor large input tiles over a large batch size and hence reduce the batch to a single image. Accordingly we use a high momentum (0.99) such that a large number of the previously seen training samples determine the update in the current optimization step.

The energy function is computed by a pixel-wise soft-max over the final feature map combined with the cross entropy loss function. The soft-max is defined as $p_k(\mathbf{x}) = \exp(a_k(\mathbf{x}))/\left(\sum_{k'=1}^{K} \exp(a_{k'}(\mathbf{x}))\right)$ where $a_k(\mathbf{x})$ denotes the activation in feature channel k at the pixel position $\mathbf{x} \in \Omega$ with $\Omega \subset \mathbb{Z}^2$. K is the number of classes and $p_k(\mathbf{x})$ is the approximated maximum-function. I.e. $p_k(\mathbf{x}) \approx 1$ for the k that has the maximum activation $a_k(\mathbf{x})$ and $p_k(\mathbf{x}) \approx 0$ for all other k. The cross entropy then penalizes at each position the deviation of $p_{\ell(\mathbf{x})}(\mathbf{x})$ from 1 using

$$E = \sum_{\mathbf{x} \in \Omega} w(\mathbf{x}) \log(p_{\ell(\mathbf{x})}(\mathbf{x})) \tag{1}$$

我们展示了关于 EM 堆栈中神经结构分割的研究成果（这是 2012 年 ISBI 启动的持续竞赛），由 Ciresan 等人[1]完成。此外，我们还展示了 2015 年 ISBI 细胞追踪挑战中光学显微镜图像中细胞分割的结果。在两个最具挑战性的二维透射光数据集上，我们以较大优势获胜。

2 网络架构

网络架构如图 1 所示。它由收缩路径（左侧）和扩张路径（右侧）组成。该收缩路径遵循卷积网络的典型架构。它由重复应用两个 3x3 卷积（无填充卷积）组成，每个卷积后接一个整流线性单元（ReLU）和一个 2x2 最大池化操作，步幅为 2 进行下采样。每下采样一步，特征通道数量翻倍。扩展路径的每一步都包括对特征图进行上采样，随后进行 2x2 卷积（“上卷”），使特征通道数量减半，再与收缩路径中相应裁剪的特征映射串接，以及两个 3x3 卷积，每个卷积后跟一个 ReLU。裁剪是因为每次卷积都会丢失边框像素。在最后一层，使用 1x1 卷积将每个 64 分量特征向量映射到所需的类别数。该网络共有 23 个卷积层。

为了实现输出分割映射的无缝铺砌（见图 2），重要的是选择输入图块大小，使所有 2x2 的最大池化操作都应用于 x 和 y 大小为偶的图层。

3 培训

输入图像及其对应的分割映射用于用 Caffe 的随机梯度下降实现训练网络[6]。由于无填充卷积，输出图像比输入小了一条恒定的边框宽度。为了最小化开销并最大化利用 GPU 内存，我们偏好大输入瓦片而非大批量，因此将批量简化为单一图像。因此，我们使用高动量（0.99），使得大量先前看到的训练样本决定当前优化步骤的更新。

能量函数通过对最终特征图进行像素级软最大值计算，结合交叉熵损失函数。软最大值定义为 p（x） = exp（a（x））/

$$\left(\sum_{k=1} \exp(a(x))\right)$$ 其中 a（x）表示

特征通道 K 在像素位置 x∈ Ω激活，Z 为Ω ⊂。K 是类别数，p（x）是近似的最大函数。即 p（x）≈k 为 1，k 激活 a（x）最大，p（x）为 0，p（x）为 0，其他 k 为≈0。交叉熵随后在每个位置惩罚 p（x）偏离 1 的次数，使用如下方式

$$E = \sum_{x\in\Omega} w(x) \text{ 对数 } (p(x)) \tag{1}$$

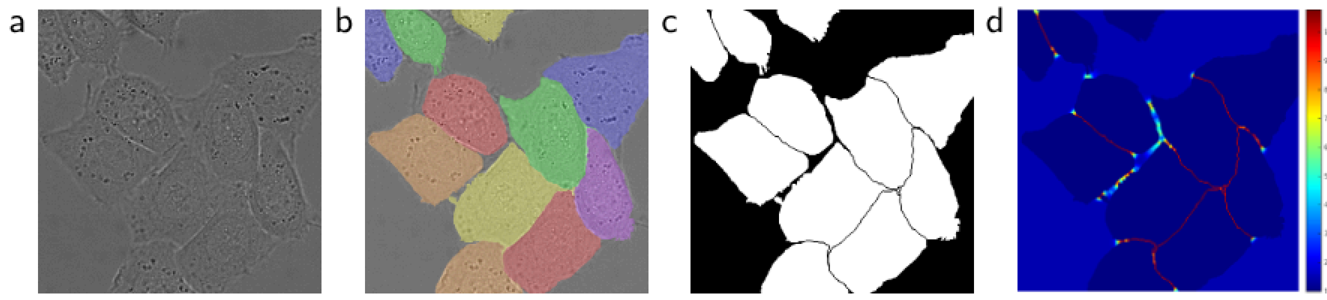


Fig. 3. HeLa cells on glass recorded with DIC (differential interference contrast) microscopy. (**a**) raw image. (**b**) overlay with ground truth segmentation. Different colors indicate different instances of the HeLa cells. (**c**) generated segmentation mask (white: foreground, black: background). (**d**) map with a pixel-wise loss weight to force the network to learn the border pixels.

where $\ell : \Omega \to \{1, \dots, K\}$ is the true label of each pixel and $w : \Omega \to \mathbb{R}$ is a weight map that we introduced to give some pixels more importance in the training.

We pre-compute the weight map for each ground truth segmentation to compensate the different frequency of pixels from a certain class in the training data set, and to force the network to learn the small separation borders that we introduce between touching cells (See Figure 3c and d).

The separation border is computed using morphological operations. The weight map is then computed as

$$w(\mathbf{x}) = w_c(\mathbf{x}) + w_0 \cdot \exp\left(-\frac{(d_1(\mathbf{x}) + d_2(\mathbf{x}))^2}{2\sigma^2}\right) \tag{2}$$

where $w_c : \Omega \to \mathbb{R}$ is the weight map to balance the class frequencies, $d_1 : \Omega \to \mathbb{R}$ denotes the distance to the border of the nearest cell and $d_2 : \Omega \to \mathbb{R}$ the distance to the border of the second nearest cell. In our experiments we set $w_0 = 10$ and $\sigma \approx 5$ pixels.

In deep networks with many convolutional layers and different paths through the network, a good initialization of the weights is extremely important. Otherwise, parts of the network might give excessive activations, while other parts never contribute. Ideally the initial weights should be adapted such that each feature map in the network has approximately unit variance. For a network with our architecture (alternating convolution and ReLU layers) this can be achieved by drawing the initial weights from a Gaussian distribution with a standard deviation of $\sqrt{2/N}$, where N denotes the number of incoming nodes of one neuron [5]. E.g. for a 3x3 convolution and 64 feature channels in the previous layer $N = 9 \cdot 64 = 576$.

3.1 Data Augmentation

Data augmentation is essential to teach the network the desired invariance and robustness properties, when only few training samples are available. In case of

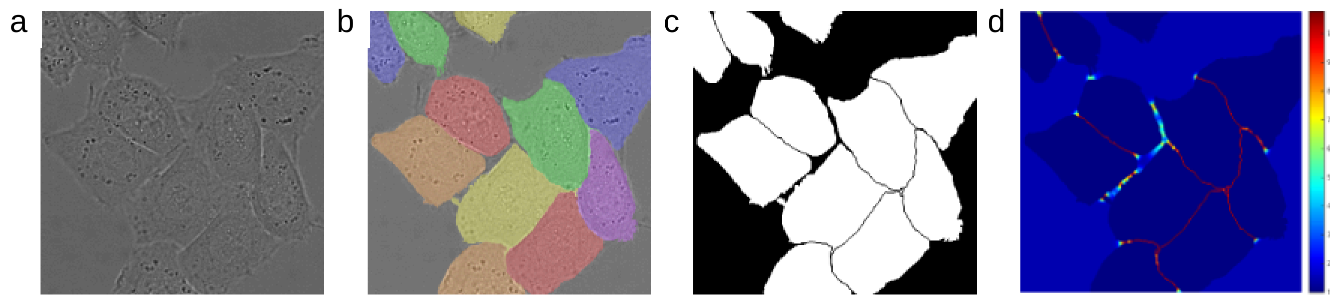


图 3。玻璃上的 HeLa 细胞通过 DIC（差分干扰对比）显微镜记录。(a) 原始图像。(b) 叠加地面真实分段。不同颜色表示 HeLa 细胞的不同实例。(c) 生成的分割遮罩（白色：前景，黑色：背景）。(d) 按像素减重的映射，迫使网络学习边界像素。

其中 '：Ω → {1，. . . ，K} 是每个像素的真实标签，w：Ω → R 是一个权重映射，我们引入该映射以增强某些像素在训练中的重要性。

我们预先计算每个基层真实分割的权重图，以补偿训练数据集中某一类别像素频率的不同，并强制网络学习我们在接触单元之间引入的小分隔边界（见图 3c 和图 d）。

分隔边界是通过形态学操作计算的。权重映射随后计算为

$$w(x) = w(x) + w\cdot \text{经验}\left(-\frac{(d(x)+d(x))}{2\sigma}\right) \tag{2}$$

其中 w：Ω →R 是用于平衡类别频率的权重映射，d：Ω →R 表示最近单元格边界的距离，d：Ω →表示第二最近单元格边界的距离。在我们的实验中，我们设 w= 10，并且σ ≈ 5 个像素。

在拥有多个卷积层和不同网络路径的深度网络中，权重的良好初始化极为重要。否则，网络的部分可能会提供过多的激活，而其他部分则从未贡献。理想情况下，初始权重应调整为使网络中的每个特征图的单位方差大致为单位。对于采用我们架构（交替卷积层和 ReLU 层）的网络，这可以通过从标准差为

$\sqrt{2/N}$，其中 N 表示单个神经元的入射节点数 [5]。例如，对于 3x3 卷积和前一层 64 个特征通道，N = 9 ·64 = 576。

3.1 数据增强

当训练样本很少时，数据增强对于教授网络所需的不变性和鲁棒性属性至关重要。在

microscopical images we primarily need shift and rotation invariance as well as robustness to deformations and gray value variations. Especially random elastic deformations of the training samples seem to be the key concept to train a segmentation network with very few annotated images. We generate smooth deformations using random displacement vectors on a coarse 3 by 3 grid. The displacements are sampled from a Gaussian distribution with 10 pixels standard deviation. Per-pixel displacements are then computed using bicubic interpolation. Drop-out layers at the end of the contracting path perform further implicit data augmentation.

4 Experiments

We demonstrate the application of the u-net to three different segmentation tasks. The first task is the segmentation of neuronal structures in electron microscopic recordings. An example of the data set and our obtained segmentation is displayed in Figure 2. We provide the full result as Supplementary Material. The data set is provided by the EM segmentation challenge [14] that was started at ISBI 2012 and is still open for new contributions. The training data is a set of 30 images (512x512 pixels) from serial section transmission electron microscopy of the Drosophila first instar larva ventral nerve cord (VNC). Each image comes with a corresponding fully annotated ground truth segmentation map for cells (white) and membranes (black). The test set is publicly available, but its segmentation maps are kept secret. An evaluation can be obtained by sending the predicted membrane probability map to the organizers. The evaluation is done by thresholding the map at 10 different levels and computation of the "warping error", the "Rand error" and the "pixel error" [14].

The u-net (averaged over 7 rotated versions of the input data) achieves without any further pre- or postprocessing a warping error of 0.0003529 (the new best score, see Table 1) and a rand-error of 0.0382.

This is significantly better than the sliding-window convolutional network result by Ciresan et al. [1], whose best submission had a warping error of 0.000420 and a rand error of 0.0504. In terms of rand error the only better performing

Table 1. Ranking on the EM segmentation challenge [14] (march 6th, 2015), sorted by warping error.

Rank	Group name	Warping Error	Rand Error	Pixel Error
	** human values **	0.000005	0.0021	0.0010
1.	u-net	**0.000353**	0.0382	0.0611
2.	DIVE-SCI	0.000355	0.0305	0.0584
3.	IDSIA [1]	0.000420	0.0504	0.0613
4.	DIVE	0.000430	0.0545	**0.0582**
⋮				
10.	IDSIA-SCI	0.000653	**0.0189**	0.1027

当训练样本较少时，显微图像主要需要移动和旋转的不变性，以及对变形和灰值变化的鲁棒性。尤其是训练样本的随机弹性变形，似乎是训练带有极少注释图像的分割网络的关键概念。我们利用随机位移矢量在粗略的 3x3 网格上生成平滑的变形。位移取自标准差为 10 像素的高斯分布。然后通过双三次插值计算每像素位移。合同路径末端的中撤层还进行隐式数据增强。

4 实验

我们演示了 u-net 在三种不同分割任务中的应用。第一个任务是在电子显微镜记录中对神经元结构进行分段。图 2 展示了数据集和我们获得的分割示例。我们提供完整结果作为补充材料。数据集由 EM 细分挑战[14]提供，该挑战始于 ISBI 2012，目前仍开放接受新贡献。训练数据是一组 30 张图像（512x512 像素），来自果蝇第一龄幼虫腹侧神经索（VNC）的串切透射电子显微镜。每张图像都附有对应的全注释基础真实切片图，分别针对细胞（白色）和膜（黑色）。测试集公开，但其分割图被保密。通过将预测的膜概率映射发送给组织者，可以获得评估。评估通过在 10 个不同层级对映射进行阈值化，并计算“扭曲误差”、“兰德误差”和“像素误差”[14]。

U-net（平均 7 个旋转输入数据版本）在无进一步预处理或后处理的情况下，扭曲误差为 0.0003529（新最佳得分，见表 1）和 0.0382 的兰德误差。

这明显优于 Ciresan 等人[1]的滑动窗口卷积网络结果，后者的最佳提交误差为 0.000420，兰德误差为 0.0504。就兰特误差而言，唯一表现更好的

表 1。EM 分割挑战[14]排名（2015 年 3 月 6 日），按扭曲误差排序。

军衔	组名称	扭曲误差	兰德误差	像素错误
	** 人类价值观 **	0.000005	0.0021	0.0010
1.	u-net	0.000353	0.0382	0.0611
2.	俯冲-SCI	0.000355	0.0305	0.0584
3.	IDSIA [1]	0.000420	0.0504	0.0613
4.	潜水	0.000430	0.0545	0.0582
⋮				
10.	IDSIA-SCI	0.000653	0.0189	0.1027

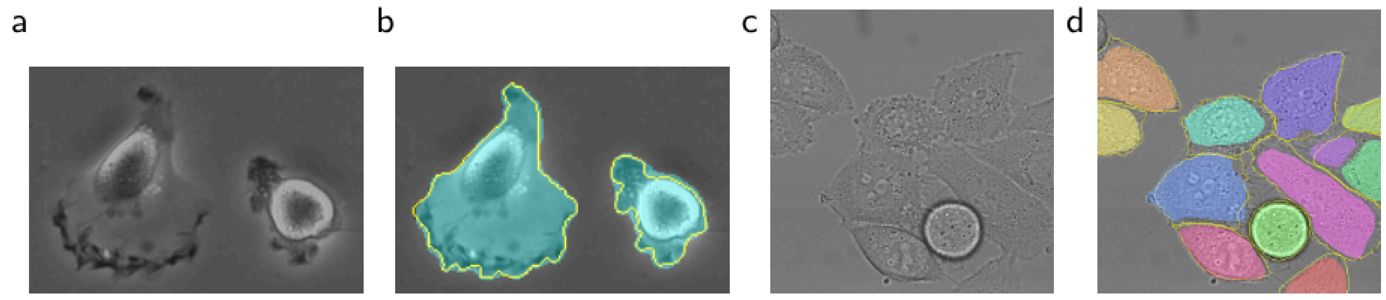


Fig. 4. Result on the ISBI cell tracking challenge. (**a**) part of an input image of the "PhC-U373" data set. (**b**) Segmentation result (cyan mask) with manual ground truth (yellow border) (**c**) input image of the "DIC-HeLa" data set. (**d**) Segmentation result (random colored masks) with manual ground truth (yellow border).

Table 2. Segmentation results (IOU) on the ISBI cell tracking challenge 2015.

Name	PhC-U373	DIC-HeLa
IMCB-SG (2014)	0.2669	0.2935
KTH-SE (2014)	0.7953	0.4607
HOUS-US (2014)	0.5323	-
second-best 2015	0.83	0.46
u-net (2015)	**0.9203**	**0.7756**

algorithms on this data set use highly data set specific post-processing methods[1] applied to the probability map of Ciresan et al. [1].

We also applied the u-net to a cell segmentation task in light microscopic images. This segmenation task is part of the ISBI cell tracking challenge 2014 and 2015 [10,13]. The first data set "PhC-U373"[2] contains Glioblastoma-astrocytoma U373 cells on a polyacrylimide substrate recorded by phase contrast microscopy (see Figure 4a,b and Supp. Material). It contains 35 partially annotated training images. Here we achieve an average IOU ("intersection over union") of 92%, which is significantly better than the second best algorithm with 83% (see Table 2). The second data set "DIC-HeLa"[3] are HeLa cells on a flat glass recorded by differential interference contrast (DIC) microscopy (see Figure 3, Figure 4c,d and Supp. Material). It contains 20 partially annotated training images. Here we achieve an average IOU of 77.5% which is significantly better than the second best algorithm with 46%.

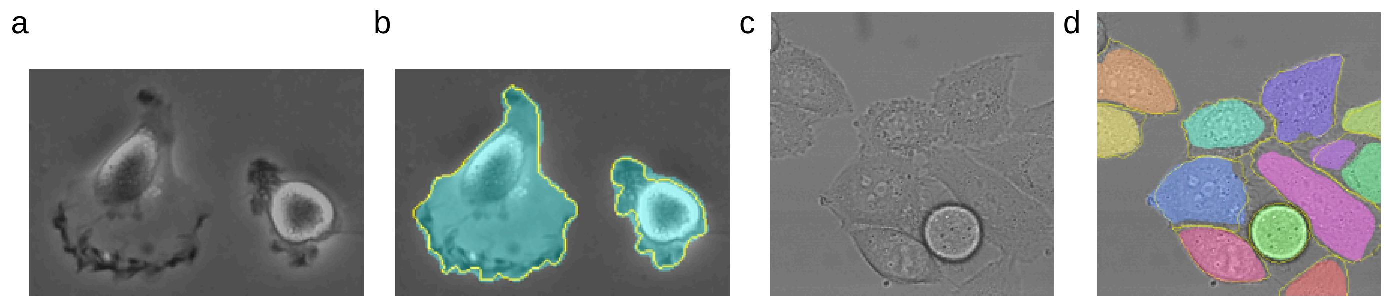
5 Conclusion

The u-net architecture achieves very good performance on very different biomedical segmentation applications. Thanks to data augmentation with elastic defor-

[1] The authors of this algorithm have submitted 78 different solutions to achieve this result.

[2] Data set provided by Dr. Sanjay Kumar. Department of Bioengineering University of California at Berkeley. Berkeley CA (USA)

[3] Data set provided by Dr. Gert van Cappellen Erasmus Medical Center. Rotterdam. The Netherlands



1027 图 4。ISBI 细胞追踪挑战的结果。(a）“PhC-U373”数据集输入图像的一部分。(b） 分割结果（青色掩模）与手动地面真实值（黄色边框） （c） “DIC-HeLa”数据集的输入图像。(d） 分割结果（随机彩色掩模）与手动地面真实值（黄色边框）。

表 2。2015 年 ISBI 小区追踪挑战的细分结果（IOU）。

名称	PhC-U373 DIC-HeLa
IMCB-SG（2014 年）0.2669 0.2935 KTH-SE	
（2014 年）0.7953 0.4607 HOUS-美国	
（2014 年）0.5323 2015 年排名第二 0.83	
0.46 净排名（2015 年）0.9203	
	0.7756

该数据集的算法采用高度数据集特异的后处理方法，应用于 Ciresan 等人的概率图[1]。

我们还将 U-net 应用于光微图像中的细胞分割任务。该隔离任务是 2014 年和 2015 年 ISBI 细胞追踪挑战的一部分[10,13]。第一个数据集“PhC-U373”包含了通过相差显微镜记录的聚丙烯酰胺基底上的胶质母细胞瘤-星形细胞瘤 U373 细胞（见图 4a、b 及补充材料）。它包含 35 张部分注释的训练图像。这里我们获得了平均 IOU（“交叉过并”）为 92%，显著优于次优算法的 83%（见表 2）。第二组“DIC-HeLa”是通过微分干涉对比（DIC）显微镜记录的平板玻璃上的 HeLa 细胞（见图 3、图 4c、d 及补充材料）。它包含 20 张部分注释的训练图像。这里我们获得了平均 77.5% 的 IOU，明显优于第二好的算法 46%。

5 结论

U-net 架构在截然不同的生物医学分段应用中实现了非常优异的性能。多亏了弹性解体的数据增强——

[1] 该算法的作者提交了 78 种不同的解决方案以实现这一结果。

[2] 数据集由 Sanjay Kumar 博士提供。加州大学伯克利分校生物工程系。美国加利福尼亚州伯克利

[3] 数据集由 Gert van Cappellen 博士 Erasmus 医疗中心提供。鹿特丹。荷兰

mations, it only needs very few annotated images and has a very reasonable training time of only 10 hours on a NVidia Titan GPU (6 GB). We provide the full Caffe[6]-based implementation and the trained networks[4]. We are sure that the u-net architecture can be applied easily to many more tasks.

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[4] U-net implementation, trained networks and supplementary material available at http://lmb.informatik.uni-freiburg.de/people/ronneber/u-net

Mations，它只需要很少的注释图片，在 NVidia Titan GPU（6 GB）上训练时间非常合理，只有 10 小时。我们提供完整的基于咖啡[6]的实现和训练有素的网络。我们相信 U-net 架构可以轻松应用于更多任务。

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[4] U-net 实施、培训网络及补充资料可于此 http://lmb.informatik.uni-freiburg.de/people/ronneber/u-net